

Rafi Al Mahmud

Adjunct Faculty Applicant, Computer Science and Engineering
Chattogram, Bangladesh | +880 1974-616402 | rafi.almahmud.007@gmail.com
Portfolio | LinkedIn | GitHub

ACADEMIC PROFILE

Computer science graduate who has built and shipped software, not just studied it. My thesis work makes object detectors more reliable in fog, and I have launched live platforms for a student magazine and a sports booking service. I want to teach introductory programming, OOP, data structures and databases, and I will bring real projects into the classroom so students see why the concepts matter.

EDUCATION

BRAC University **Sep 2022 – Sep 2026**
B.Sc. in Computer Science and Engineering | CGPA: **3.57 / 4.00** *Dhaka, Bangladesh*

Islamia Degree College **2019 – 2021**
Higher Secondary Certificate (HSC) | GPA: **5.00 / 5.00** *Chattogram, Bangladesh*

RESEARCH EXPERIENCE

WASE: Weather-Aware Surgical Evaluation for Fog-Robust Object Detection | *PyTorch, YOLOv8-Nano* **2026**

- B.Sc. thesis, BRAC University. Co-developed a self-training framework for fog-robust object detection that uses confidence-based pseudo-labels and quality filtering, so no extra labelled data from the target domain is needed.
- Measured an 8–12% gain in mAP@50 over the baseline, with inference around 17 ms in our test setup. Also looked at AURA, a reinforcement-learning approach to image restoration.
- Manuscript in preparation for submission to ICCV.

TEACHING INTERESTS

- **Courses:** Programming Fundamentals, Object-Oriented Programming, Data Structures and Algorithms, Database Management Systems, Web Development, Software Engineering, and an introduction to AI and computer vision.
- **Approach:** Work through examples in class, give students coding exercises they can try right away, publish grading rubrics before an assignment is due, and keep time open for consultation.

PROJECTS AND APPLIED EXPERIENCE

MangoStudents: Student Digital Magazine | *Laravel, PHP, MySQL* | mangostudent.com **2025 – 2026**

- Primary developer of a live magazine site for students. Content lives in a database, and editors publish through an admin panel without touching code.

SportEvo: Sports Facility Booking Platform | *React, TypeScript, Laravel, MySQL* | sportevo.games **2026**

- Built a live platform for finding sports complexes and requesting bookings. It handles slots per field, role-based access, overlap checks on bookings, owner approvals and a commission ledger.

MindWell: Employee Wellness Platform | *FastAPI, PostgreSQL, React, Docker* | GitHub **2025 – 2026**

- Wrote the REST APIs, database models, JWT authentication and role-based access control, and packaged everything in Docker so it runs the same on any machine.

Adverse Weather Image Classifier | *PyTorch, MobileNetV3, ONNX* | GitHub **2024**

- Trained a four-class weather classifier, checked it with precision, recall, F1 and confusion matrices, and tried ONNX export for CPU inference.

FREELANCE AND UNIVERSITY SERVICE

- **Freelance, Upwork and Fiverr:** Took client requests, worked out requirements, built the features and revised them after feedback.
- **BRAC University Cultural Club, Associate Sub-Body Member (2024 – 2026):** Worked with club leadership on the club website, including member registration and records for a club of more than 500 members.

SKILLS AND CERTIFICATION

- **Languages:** Python, PHP, Java, JavaScript/TypeScript, SQL **Databases:** PostgreSQL, MySQL
- **Development:** FastAPI, Laravel, React, REST APIs, Docker, Git/GitHub, Postman **ML:** PyTorch, TensorFlow, YOLO, ONNX
- **Certification:** Associate Data Analyst, DataCamp **Spoken:** Bengali, English